

Retail Statistical Analysis Report

Tool: IBM SPSS Statistics

Project Overview

Objective: This statistical analysis aims to identify the operational factors associated with retail risk and evaluate whether inventory classifications influence operational risk. Exploratory and inferential statistical techniques were applied to support business decision-making and identify significant predictors of retail risk.

Data Exploration and Assumption Testing

Descriptive Statistics

Descriptive statistics were computed to summarize the distribution of the retail variables before inferential analysis. The dataset consisted of 15,000 retail observations containing pricing, inventory, promotional, customer, and operational variables.

Normality Assessment

Normality was evaluated using the Kolmogorov–Smirnov test together with graphical diagnostics, including histograms and Q–Q plots. Most variables exhibited statistically significant departures from normality ($p < .001$), while stock level did not significantly deviate from normality ($p = .113$). Given the large sample size, these results were expected because formal normality tests are highly sensitive to minor deviations. The graphical assessments indicated only slight departures from normality, supporting the use of parametric statistical methods.

Outlier Assessment

Boxplots and extreme value diagnostics identified several high and low observations across the continuous variables. These observations fell within plausible retail operating ranges and were therefore retained to preserve the natural variability of the dataset.

Pearson Correlation Analysis

Pearson correlation analysis showed that all variables were significantly associated with risk score ($p < .001$). Return rate demonstrated the strongest positive relationship with retail risk ($r = .608$), whereas customer sentiment exhibited the strongest negative relationship ($r = -.561$). Competitor price ($r = -.206$) and weather index ($r = -.238$) also showed negative relationships with operational risk. Other variables displayed relatively weak correlations. This means customer-related variables demonstrated stronger relationships with retail risk than pricing or inventory variables, suggesting that customer experience and product returns are key operational considerations.

Multiple Linear Regression Analysis

Multicollinearity Assessment

Initial regression diagnostics indicated severe multicollinearity between price and competitor price, as well as moderate multicollinearity between discount and promotion intensity. Consequently, price and discount were removed from the final regression model. The remaining predictors exhibited acceptable VIF values ranging from 1.593 to 4.628, indicating no substantial multicollinearity.

Model Performance

The regression model was statistically significant, $F(8, 14991) = 1435.211$, $p < .001$. The model achieved $R = .659$, $R^2 = .434$, and Adjusted $R^2 = .433$, indicating that approximately 43.4% of the variation in retail risk score was explained by the operational variables included in the model.

Significant Predictors

Predictor	Standardized β	p-value	Interpretation
Customer Sentiment	-0.376	< .001	Strongest negative predictor of retail risk
Return Rate	0.297	< .001	Strongest positive predictor of retail risk
Competitor Price	-0.152	< .001	Higher competitor prices are associated with lower retail risk
Stock Level	-0.094	< .001	Higher stock levels are associated with lower retail risk

Footfall, advertising spend, weather index, and promotion intensity were not statistically significant predictors after controlling for the remaining variables.

One-Way ANOVA

The ANOVA indicated a statistically significant difference in mean risk scores among stock status groups, $F(2, 14997) = 20.632$, $p < .001$. Tukey's HSD post hoc test revealed significant differences between all pairwise stock status categories. The highest mean risk score was observed in Low Stock products, followed by High Stock, while Medium Stock products exhibited the lowest average risk. Furthermore, the results suggest that inventory classification is associated with differences in operational risk, emphasizing the importance of maintaining balanced inventory levels.

Chi-Square Test of Independence

A Chi-square test of independence was conducted to determine whether stock status is associated with retail risk category. The analysis revealed a statistically significant association between the two categorical variables, $\chi^2(4, N = 15,000) = 55.854$, $p < .001$, indicating that the distribution of risk categories varies across stock status groups. Examination of the cross-tabulation showed that the majority of products in each stock status category were classified as Low Risk, while High Risk products comprised only a small proportion of the dataset.

To assess the practical significance of the relationship, Cramer's V was computed and yielded a value of 0.043, indicating a very weak association between stock status and risk category. Although the relationship was statistically significant,

the small effect size suggests that stock status alone has limited influence on retail risk classification. This implies that operational risk is more likely driven by multiple interacting factors, including customer sentiment, return rate, competitor pricing, and inventory-related conditions identified in the regression analysis. Consequently, stock status should be considered as one component of a broader retail risk management strategy rather than as a standalone determinant of operational risk.

Key Findings

- Customer sentiment was the strongest negative predictor of retail risk.
- Return rate was the strongest positive predictor of retail risk.
- Competitor price and stock level significantly reduced retail risk.
- Inventory groups differed significantly in their average risk scores.
- Stock status and risk category were statistically associated, although the relationship was weak.
- The regression model explained 43.4% of the variation in retail risk score.

Business Recommendations

- Prioritize initiatives that improve customer satisfaction to reduce operational risk.
- Monitor and reduce product return rates through quality assurance and return policy evaluation.
- Maintain appropriate inventory levels to minimize operational disruptions.
- Track competitor pricing to support strategic pricing decisions.
- Consider additional operational and market variables in future analyses to improve predictive performance.